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(54) **LIGHTNING INDUCTION SYSTEM AND METHOD THEREFOR**

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B64U 101/00 (2023.01)

(52) **U.S. Cl.**

CPC ***B64D 45/02*** (2013.01); *B64U 2101/00* (2023.01)

(58) **Field of Classification Search**

CPC B64D 45/02; B64U 2101/00

See application file for complete search history.

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(57) **ABSTRACT**

A lightning induction system includes a plurality of flight vehicles **3** that fly between a lightning originating point **1** in the sky where lightning is generated and a ground surface **2**. The flight vehicles **3** each include: a flight unit **10** that generates lift force and propulsion force; a conductor cable **20** that is engaged with the flight unit **10**, extends downward, and has a predetermined length; and a weight unit **30** connected to the lower end of the conductor cable **20**. The flight unit **10** is disposed in a Faraday cage **11**. The Faraday cage **11** has a sphere shape, and the flight unit **10** is fixed by a support pillar **12** extending downward from the inner side of the top of the sphere.

6 Claims, 5 Drawing Sheets

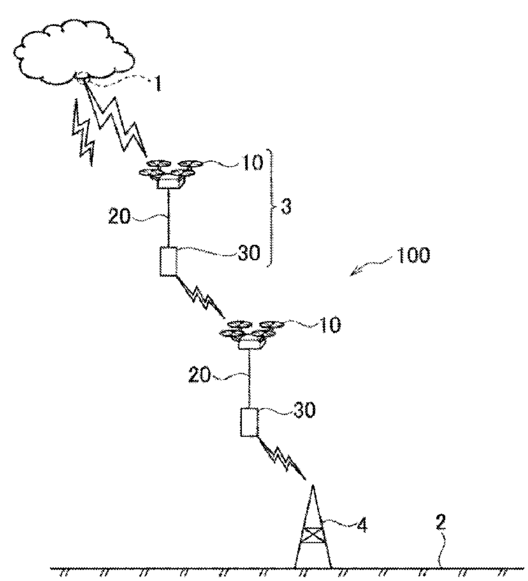

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Fig. 1

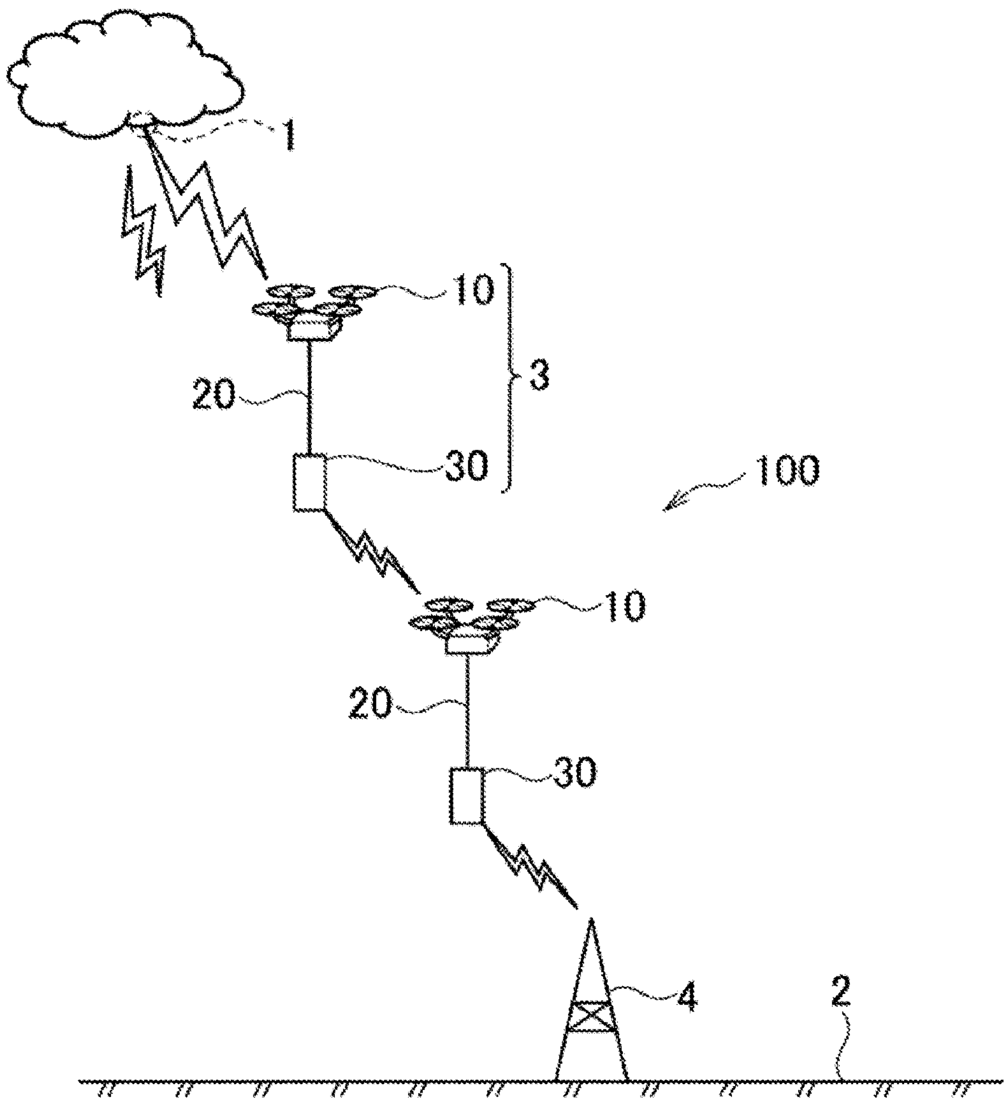


Fig. 2

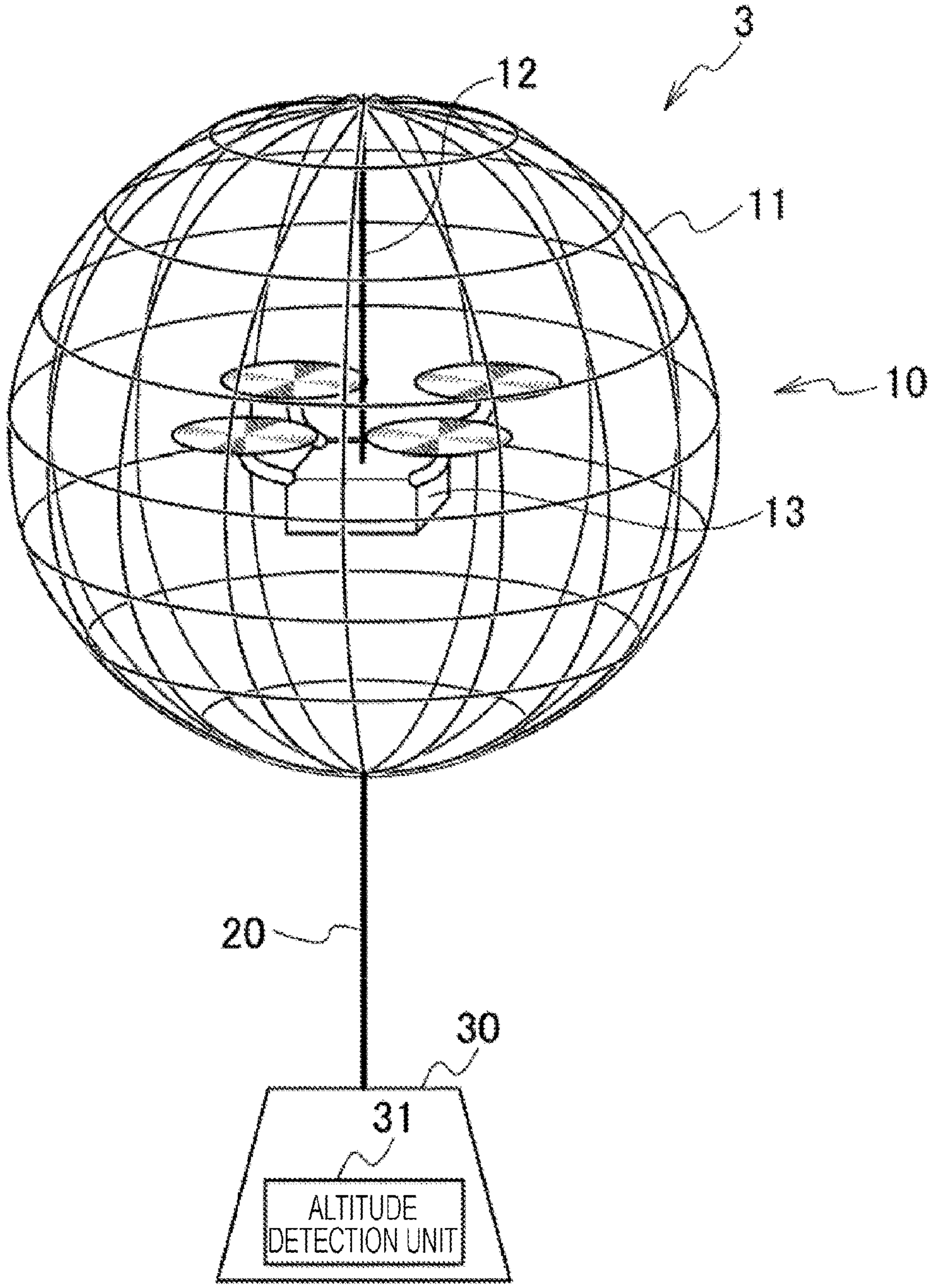


Fig. 3

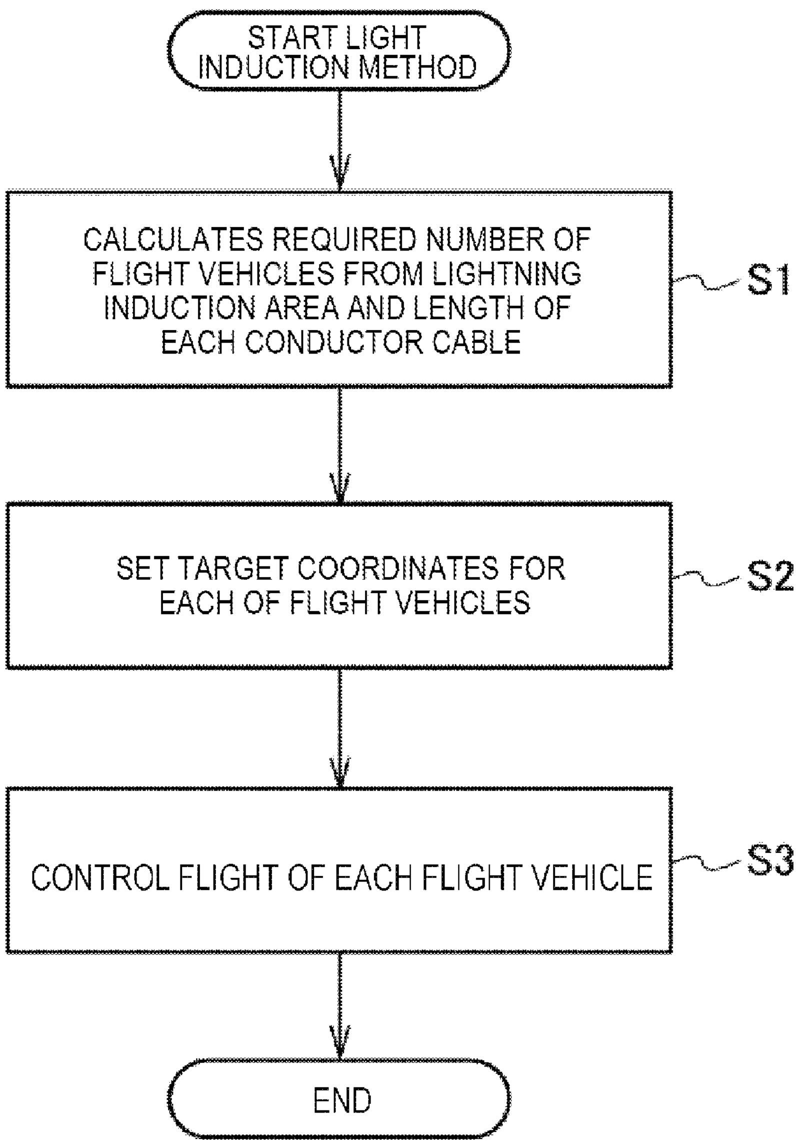

START LIGHT
INDUCTION METHOD
CALCULATES REQUIRED NUMBER OF FLIGHT VEHICLES FROM LIGHTNING INDUCTION AREA AND LENGTH OF EACH CONDUCTOR CABLE
S1
SET TARGET COORDINATES FOR EACH OF FLIGHT VEHICLES
S2
CONTROL FLIGHT OF EACH FLIGHT VEHICLE
S3
END

Fig. 4

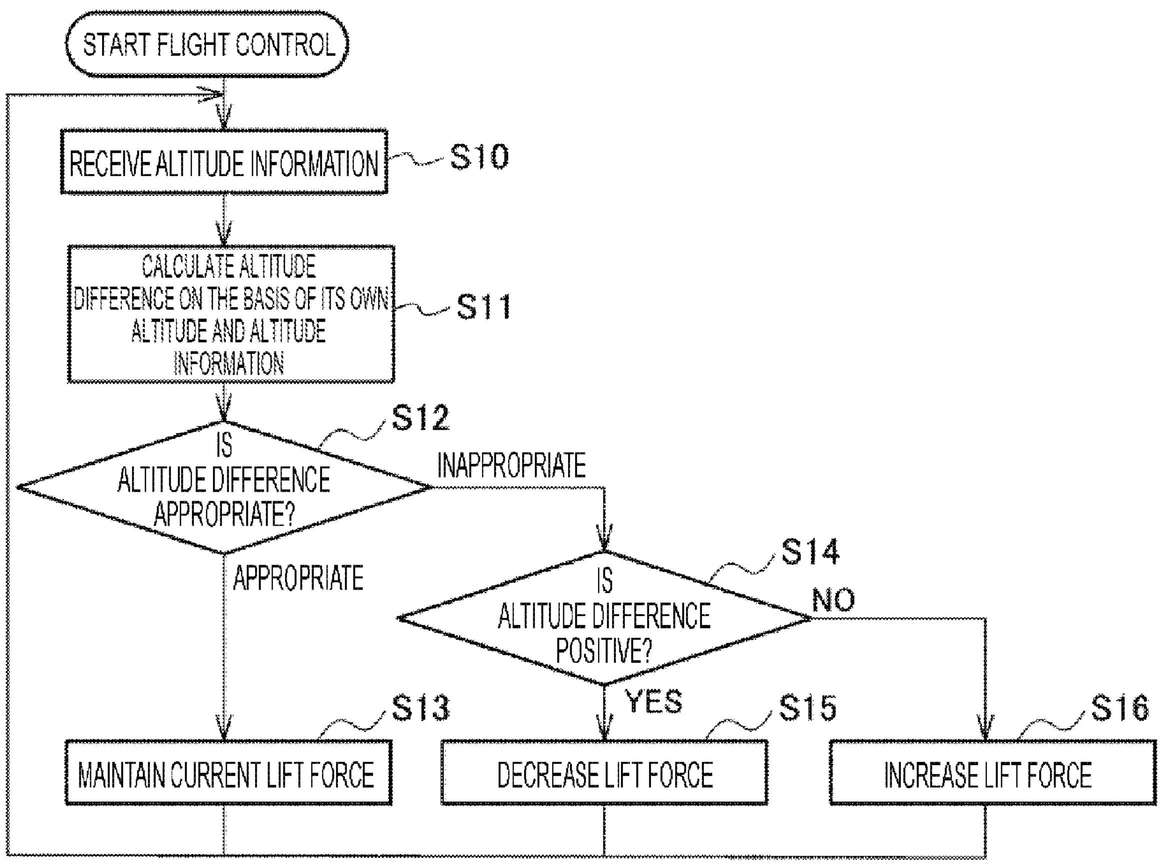
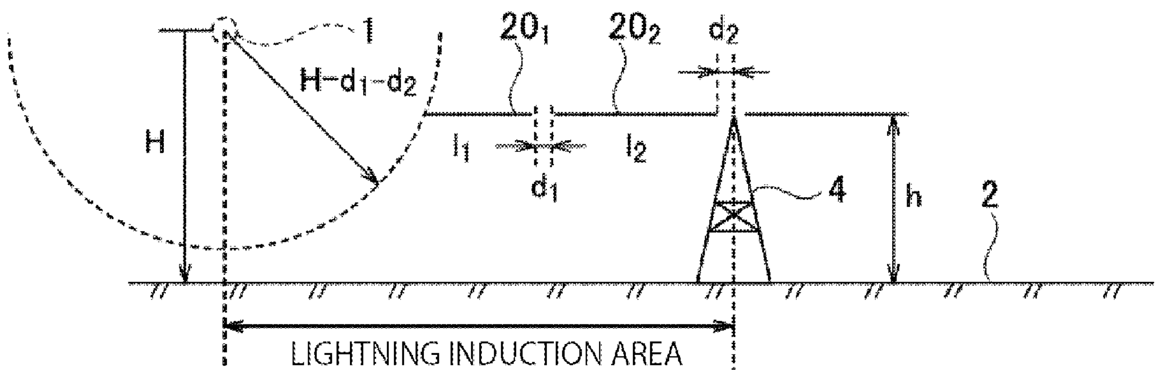

START FLIGHT CONTROL
RECEIVE ALTITUDE INFORMATION
S10
CALCULATE ALTITUDE DIFFERENCE ON THE BASIS OF ITS OWN ALTITUDE AND ALTITUDE INFORMATION
S11
IS ALTITUDE DIFFERENCE APPROPRIATE?
S12
INAPPROPRIATE
APPROPRIATE
IS ALTITUDE DIFFERENCE POSITIVE?
S14
NO
YES
MAINTAIN CURRENT LIFT FORCE
S13
DECREASE LIFT FORCE
S15
INCREASE LIFT FORCE
S16

Fig. 5



LIGHTNING INDUCTION SYSTEM AND METHOD THEREFOR

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a National Stage application under 35 U.S.C. § 371 of International Application No. PCT/JP2021/008121, having an International Filing Date of Mar. 3, 2021. The disclosure of the prior application is considered part of the disclosure of this application, and is incorporated by reference in its entirety into this application.

TECHNICAL FIELD

The present invention relates to a lightning induction system and a method therefor.

BACKGROUND ART

Lightning strikes anywhere, depending on the position of the cumulonimbus cloud that is the source of lightning. For example, accidents are inevitable when lightning strikes substations of a transportation system and disrupts train operations. Also, human damage still occurs several times a year, and death accidents are reported.

Studies have been made to prevent such damage to be caused by a lightning strike. For example, Non Patent Literature 1 discloses a technology for flying a drone having a conductor cable engaged between the lightning originating point in the sky where the lightning is generated and the ground surface, and thus inducing the lightning to a desired position (lightning induction).

CITATION LIST

Non Patent Literature

Non Patent Literature 1: [retrieved on Feb. 12, 2021], Internet <URL: https://monoist.atmarkit.co.jp/mn/articles/2011/20/new s052.html>

SUMMARY OF INVENTION

Technical Problem

With the technology disclosed in Non Patent Literature 1, however, it is necessary to fly the drone high to capture lightning in a wide area. Flying the drone high leads to an increase in the length of the conductor cable between the drone and the lightning strike target where lightning is struck.

If the conductor cable is made longer, the load on the drone increases due to the weight of the conductor cable and the wind pressure, and the risk of accidents due to contact between the conductor cable and a ground object other than the lightning strike target becomes higher.

The present invention has been made in view of the above problems, and aims to provide a lightning induction system and a method therefor that can shorten the length of conductor cables, reduce the load on drones, and lower the risk of accidents.

Solution to Problem

A lightning induction system according to one mode of the present invention is a lightning induction system that includes a plurality of flight vehicles that fly between a lightning originating point in the sky where lightning is generated and a ground surface. In the lightning induction system, the flight vehicles each include: a flight unit that generates lift force and propulsion force; a conductor cable that is engaged with the flight unit, extends downward, and has a predetermined length; and a weight unit connected to a lower end of the conductor cable.

Also, a lightning induction method according to one mode of the present invention is a lightning induction method for inducing energy of lightning striking a ground surface from a lightning originating point in the sky where lightning is generated, to the ground surface via a plurality of flight vehicles. The flight vehicles each include: a flight unit that generates lift force and propulsion force; a conductor cable that is engaged with the flight unit, extends downward, and has a predetermined length; and a weight unit connected to a lower end of the conductor cable.

Advantageous Effects of Invention

According to the present invention, it is possible to provide a lightning induction system and a method therefor that can shorten the length of conductor cables, reduce the load on flight units, and thus lower the risk of accidents.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a diagram schematically illustrating an example in which a lightning induction system according to an embodiment of the present invention induces lightning.

FIG. **2** is a diagram illustrating a flight unit, a conductor cable, and a weight unit that constitute a flight vehicle illustrated in FIG. **1**.

FIG. **3** is a flowchart illustrating the processing procedures of a lightning induction method according to an embodiment of the present invention.

FIG. **4** is a flowchart illustrating some of the processing procedures to be carried out by a flight control unit illustrated in FIG. **2**.

FIG. **5** is a schematic diagram illustrating an example layout of conductor cables of a lightning induction system according to a modified example.

DESCRIPTION OF EMBODIMENTS

In the following, an embodiment of the present invention will be described with reference to the drawings. Throughout the drawings, the same components are denoted by the same reference numerals, and the description thereof will not be repeated.

FIG. **1** is a diagram schematically illustrating an example in which a lightning induction system according to an embodiment of the present invention induces lightning. A lightning induction system **100** illustrated in FIG. **1** includes a plurality of flight vehicles **3**. FIG. **1** illustrates an example in which two flight vehicles **3** are flown between a lightning originating point **1** in a thundercloud and a ground surface **2**, and lightning is induced to a lightning strike target that is a steel tower **4**, for example.

The flight vehicle **3** includes a flight unit **10**, a conductor cable **20**, and a weight unit **30**.

The flight unit **10** is a wireless flight vehicle generally called a drone, and normally flies by remote control performed by a drone pilot (not illustrated) on the ground. The flight unit **10** and the remote controller (not illustrated) being operated by the drone pilot are wirelessly connected. The two flight units **10** may be remotely controlled by one drone pilot, or may be remotely controlled by different drone pilots.

Note that the flight unit **10** may not be a wireless flight vehicle. A manned flight vehicle such as a helicopter may be used. In that case, a drone pilot to perform remote control is not necessary either.

Alternatively, the flight vehicles **3** may operate on autopilot. In that case, the required number of flight vehicles **3**, the length of each conductor cable **20**, and the target coordinates of each flight vehicle **3**, which are the information in securing the necessary lightning induction area (=the area to be protected), are calculated in advance, and each flight vehicle **3** is informed of the target coordinate, and thus each flight vehicle **3** operates on autopilot.

One end of the conductor cable **20** is engaged with the flight unit **10**, and induces the energy of lightning that has struck the flight unit **10** downward. The current of lightning reaches several hundreds of thousands of ampere (A), but the generation time thereof is about $\frac{1}{10{,}000}$ of a second to about $\frac{1}{1{,}000}$ of a second.

Accordingly, even when a power cord having a cross-sectional area of 8 mm^2 and an allowable current of about 61 A is used as the conductor cable **20**, burning does not occur. The length of the conductor cable **20** is about 100 m, for example.

The weight unit **30** is engaged with the other end of the conductor cable **20**. The weight unit **30** generates tension in a vertically downward direction in the conductor cable **20**, and stabilizes the positions of the flight unit **10** and the conductor cable **20**.

The distances between the flight vehicles **3** are appropriately controlled, and thus a lightning induction path with a low impedance is formed between the lightning originating point **1** and the lightning strike target (the steel tower **4** in this example). Since the lightning induction path is formed with a path having the lowest impedance, a predetermined distance may be allowed between an upper weight unit **30** and a lower flight unit **10**.

As described above, the lightning induction system **100** according to the present embodiment is a lightning induction system that includes a plurality of flight vehicles **3** that fly between the lightning originating point **1** in the sky where lightning is generated and the ground surface **2**. The flight vehicles **3** each include: the flight unit **10** that generates lift force and propulsion force; the conductor cable **20** that is engaged with the flight unit **10**, extends downward, and has a predetermined length; and the weight unit **30** connected to the lower end of the conductor cable **20**. With this configuration, it is possible to provide the lightning induction system **100** that can shorten the length of the conductor cable **20**, and lower the risk of accidents.

According to the lightning induction system **100**, the conductor cable **20** can be shortened to reduce the load on the drone, and the risk of entanglement of the conductor cable **20** and contact with ground equipment can be lowered. Furthermore, since there is no need to connect the lightning strike target and the conductor cable **20**, a facility having a great height can be easily set as the lightning strike target. Thus, lightning can be more safely induced.

(Flight Vehicle)

FIG. **2** is a diagram illustrating the flight unit **10**, the conductor cable **20**, and the weight unit **30** that constitute the flight vehicle **3**.

The flight unit **10** may be disposed in a Faraday cage **11**. The flight unit **10** in this case is disposed in the Faraday cage **11** that is a spherical basket, for example.

The Faraday cage **11** is a space surrounded by conductors, or a conductor basket or vessel to be used to create such a space. Since any electric line of force cannot enter the inside surrounded by the conductors, external electric fields are blocked, and potentials inside are all equal.

As illustrated in FIG. **2**, the flight unit **10** is fixed by one support pillar **12** extending downward from the inner side of the top of the Faraday cage **11**. The flight unit **10** is fixed inside the Faraday cage **11** at a predetermined distance from the inner side so as not to disturb the flight.

With this arrangement, the flight unit **10** and the Faraday cage **11** are connected at one point. Thus, the flight unit **10** is not affected by the energy of lightning (lightning strike). Alternatively, the influence of lightning strike can be reduced.

The energy of lightning that has struck the Faraday cage **11** is discharged to the Faraday cage **11** of another flight vehicle **3** via the conductor cable **20** and the weight unit **30**.

The flight unit **10** includes a flight control unit **13**. The flight control unit **13** receives an operation signal from the drone pilot, and controls generation of lift force and propulsion force in the flight unit **10**. The flight control unit **13** also has a function of measuring its own altitude.

The flight control unit **13** can be realized by a computer that includes a ROM, a RAM and a CPU, for example. In that case, the processing contents are written in a program.

The weight unit **30** may include an altitude detection unit **31**. The altitude detection unit **31** and the flight control unit **13** may measure altitude, using either an altimeter formed with a pressure sensor or a GPS. Note that, in FIG. **2**, the power supply and the like that activate the altitude detection unit **31** are not illustrated.

(Lightning Induction Method)

FIG. **3** is a flowchart illustrating the processing procedures of a lightning induction method implemented with the lightning induction system **100** according to the embodiment of the present invention. The lightning induction method will be described with reference to FIG. **3**.

First, the administrator using the lightning induction system **100** calculates the required number of flight vehicles **3** from the lightning induction area, and the length of the conductor cable **20** engaged with each of the flight vehicles **3**, so as to secure the area (the lightning induction area) to be protected from lightning (step S**1**). The calculation may be performed by a control unit (not illustrated) that controls the lightning induction system **100**, or may be performed by the administrator.

Next, the administrator sets target coordinates for each of the flight vehicles **3** (step S**2**). The target coordinates are set in three dimensions.

Next, the flight of each flight vehicle **3** is controlled (step S**3**). The flight control may be performed by a drone pilot, or may be automatically controlled.

Also, the altitude detection unit **31** of a first flight unit **10** that flies in the sky may cooperate with the flight control unit **13** of a second flight unit **10** that flies below the first flight unit **10**. That is, the drone pilot operates only the first flight unit **10**. The second flight unit **10** then flies by following, like a duck family (marching), the weight unit **30** engaged with the first flight unit **10**.

FIG. **4** is a flowchart illustrating some of the processing procedures to be carried out by the flight control unit **13**. Referring to FIG. **4**, an example of flight control to be performed in a case where a plurality of flight vehicles **3** is flown will be described.

The second flight unit **10** is flying below the first flight unit **10** in this example. The flight control unit **13** (hereinafter, referred to as the flight control unit $\mathbf{13}_2$) of the second flight unit **10** receives altitude information detected by the altitude detection unit **31** (hereinafter, referred to as the altitude detection unit $\mathbf{31}_1$) of the weight unit **30** engaged with the first flight unit **10** (hereinafter, referred to as the flight unit $\mathbf{10}_1$) (step S**10**). The subscript "$_1$" represents each functional component of the first flight unit $\mathbf{10}_1$, and the subscript "$_2$" represents each functional component of the second flight unit $\mathbf{10}_2$.

The flight control unit $\mathbf{13}_2$ calculates the altitude difference from the weight unit $\mathbf{30}_1$ on the basis of its own altitude information and the altitude information received from the weight unit $\mathbf{30}_1$ of the first flight unit $\mathbf{10}_1$ (step S**11**).

Next, the flight control unit $\mathbf{13}_2$ determines whether the altitude difference is appropriate (step S**12**). An appropriate altitude difference is an altitude difference within 10 m, for example.

If the altitude difference is determined to be appropriate ("appropriate" in step S**12**), the flight control unit $\mathbf{13}_2$ maintains the current lift force (step S**13**).

If the altitude difference is determined to be inappropriate ("inappropriate" in step S**12**), the flight control unit $\mathbf{13}_2$ then determines whether the altitude difference is positive (step S**14**). An altitude difference being positive means that the second flight unit $\mathbf{10}_2$ is close to the weight unit $\mathbf{30}_1$.

If the altitude difference is positive (YES in step S**14**), the flight control unit $\mathbf{13}_2$ decreases the lift force (step S**15**). If the altitude difference is negative (NO in step S**14**), the flight control unit $\mathbf{13}_2$ increases the lift force (step S**16**).

The above processing in steps S**10** to S**16** is repeated, and thus the weight unit $\mathbf{30}_1$ in the sky and the flight unit $\mathbf{10}_2$ flying below the weight unit $\mathbf{30}_1$ maintain an appropriate altitude difference while flying.

As described above, the lightning induction system **100** may be configured such that the weight unit $\mathbf{30}_1$ engaged with the first flight unit $\mathbf{10}_1$ includes the altitude detection unit $\mathbf{31}_1$ that measures the altitude and transmits the measured altitude information to the other flight unit $\mathbf{10}_2$, and the second flight unit $\mathbf{10}_2$ includes the flight control unit $\mathbf{13}_2$ that receives the altitude information and controls the lift force so as to maintain a predetermined altitude difference from the weight unit $\mathbf{30}_1$. Thus, the lightning induction system **100** can be easily operated.

MODIFIED EXAMPLE

Note that a modified example in which the weight unit **30** is formed with the flight unit **10** is also possible. According to a lightning induction system **100** of this modified example, lightning can be induced in a horizontal direction.

FIG. **5** is a schematic diagram illustrating a situation in which the lightning induction system **100** of the modified example is inducing lightning in a horizontal direction. In the lightning induction system **100** of the modified example, both ends of conductor cables **20** are connected to flight units **10**. In FIG. **5**, the flight units **10** are not illustrated.

According to the lightning induction system **100** of the modified example, conductor cables $\mathbf{20}_1$ and $\mathbf{20}_2$ can be arranged in a horizontal direction at predetermined intervals d_1 and d_2, as illustrated in FIG. **5**. According to the modified example, it is possible to induce lightning from a lightning originating point **1** that is located farther than the original lightning induction area of the steel tower **4**.

That is, the horizontal distance from the lightning originating point **1** to the steel tower **4** (lightning strike target) can be extended as shown in the following expression.

[Mathematical Expression 1]

$$\text{Horizontal length of lightning induction area} < \sqrt{(H-d_1-d_2)^2-(H-h)^2}+l_1+l_2+d_1+d_2 \tag{1}$$

Here, l_1 represents the length of the conductor cable $\mathbf{20}_1$, and l_2 represents the length of the conductor cable $\mathbf{20}_2$.

As described above, according to the lightning induction system **100**, one or more flight vehicles **3** including the conductor cables **20** for relay fly between the flight vehicle $\mathbf{3}_1$ for capturing lightning which includes the conductor cable $\mathbf{20}_1$ and the lightning strike target (the steel tower **4**, for example) so as to form a low-impedance path via the respective conductor cables **20**. Accordingly, the conducting wire lengths of the conductor cables **20** can be shortened, and thus the load due to the lengths of the conductor cables **20** and the risk of contact with a ground object other than the lightning strike target can be reduced.

In the example described in the above embodiment, the number of flight vehicles **3** is two, but the present invention is not limited to this example. The number of flight vehicles **3** may be n (n being a natural number). Further, the shape of the Faraday cage **11** is not necessarily a sphere. Moreover, the Faraday cage **11** is not essential.

As described above, the present invention of course includes various embodiments and the like not described herein. Therefore, the technical scope of the present invention is defined only by matters to specify the invention according to the scope of claims pertinent based on the foregoing description.

REFERENCE SIGNS LIST

1 lightning originating point
2 ground surface
3 flight vehicle
4 lightning strike target (steel tower)
10 flight unit
11 Faraday cage
12 support pillar
13 flight control unit
20 conductor cable
30 weight unit
31 altitude detection unit
100 lightning induction system

The invention claimed is:

1. A lightning induction system comprising a plurality of flight vehicles that fly between a lightning originating point in the sky where lightning is generated and a ground surface, wherein the flight vehicles each include:
- a flight unit that generates lift force and propulsion force;
- a conductor cable that is engaged with the flight unit, extends downward, and has a predetermined length; and
- a weight unit connected to a lower end of the conductor cable, wherein a conductor cable of an upper flight vehicle is configured to induce an energy of the lightning that has struck a flight unit of the upper flight vehicle to a lower flight vehicle through the weight unit.

2. The lightning induction system according to claim 1, wherein

the weight unit engaged with a first flight unit of flight units includes an altitude detection unit that measures an altitude and transmits the measured altitude to a second flight unit, and

wherein the second flight unit includes a flight control unit that receives the measured altitude and controls the lift force so as to maintain a predetermined altitude difference from the weight unit of the first flight unit,

wherein the first flight unit and the second flight unit form a relay of lightning induction path to the ground.

3. The lightning induction system of claim 1, further comprising:

a first conductor cable that is engaged with a first flight unit of flight units, extending in a horizontal direction; and

a second conductor cable that is engaged with a second flight unit of the flight units, extending in a horizontal direction,

wherein the first conductor cable and the second conductor cable are oriented in a substantially horizontal direction between the first flight unit and the second flight unit, wherein the first flight unit and the second flight unit form a horizontal relay of lightning induction path.

4. The lightning induction system according to claim 1, wherein the flight unit is disposed in a Faraday cage.

5. The lightning induction system according to claim 4, wherein

the Faraday cage has a sphere shape, and

the flight unit is fixed by a support pillar extending downward from an inner side of a top of the Faraday cage.

6. A lightning induction method for inducing energy of lightning striking a ground surface from a lightning originating point in the sky where lightning is generated, to the ground surface via a plurality of flight vehicles,

wherein the flight vehicles each include:

a flight unit that generates lift force and propulsion force;

a conductor cable that is engaged with the flight unit, extends downward, and has a predetermined length; and

a weight unit connected to a lower end of the conductor cable, wherein a conductor cable of an upper flight vehicle is configured to induce an energy of the lightning that has struck a flight unit of the upper flight vehicle to a lower flight vehicle through the weight unit.

* * * * *